

NYU Tandon School of Engineering
Spring 2026, ECE 9953 (Dr. Karri) - Independent Project
Using LLMs for UVM Testbench Generation
Project Proposal

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Project Statement:

UVM testbench development is repetitive, time-intensive, and highly pattern-driven. This project explores whether LLMs can act as verification co-pilots, accelerating testbench creation while ensuring correctness, coverage metrics, and UVM best practices.

Objectives:

- Build a golden reference database of UVM testbenches for standard digital blocks and protocols
- Design a script to auto-generate UVM testbenches using LLM APIs
- Quantitatively and qualitatively benchmark LLM outputs vs golden testbenches
- Iterate using feedback loops to improve structural, functional, and stylistic correctness

Components Chosen and Proposed Timeline:

Week	Week Start	Task
0	1/26	Ramp-Up on UVM Concepts and Structure
1	2/2	Multiplexer, Decoder/Encoder, Comparator
2	2/9	Adder, ALU, Barrel Shifter
3	2/16	Counter, Shift Register
4	2/23	FSM Controller, Synchronous FIFO
5	3/2	APB Slave, Interrupt Controller
6	3/9	AHB-Lite Slave, Reset Controller
7	3/16	AXI Source/Sink, UART TX/RX
8	3/23	Async FIFO, DMA Engine
9	3/30	Development of Script Using LLM APIs
10	4/6	Development of Script Using LLM APIs
11	4/13	Deployment of Script to Generate UVM Testbenches
12	4/20	Feedback Loops to Improve Results
13	4/27	Buffer Week for Final Changes
14	5/4	Project Report Submission and Presentation

Tentative LLMs Chosen:

GPT-5 mini

Claude Sonnet 4.5

Gemini 2 Pro

Evaluation:

Evaluation will be done comparing

- Structural Completeness
- Compile and Runtime Success
- Functional Coverage and Correctness
- Feedback Loops Required